```
#Amritansh Singh 23070521013
!pip install tensorflow scikit-learn pandas matplotlib seaborn
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.20.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
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```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

```
data = load_breast_cancer()

X = data.data
y = data.target

print(X.shape)
print(y.shape)
```

```
(569, 30)
(569,)
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
model = Sequential()

model.add(Dense(64, activation='relu', input_shape=(30,)))

model.add(Dense(32, activation='relu'))

model.add(Dense(1, activation='sigmoid'))
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

```
history = model.fit(
    X_train,
    y_train,
    epochs=30,
    batch_size=16,
    validation_split=0.2,
    verbose=1
)
```

```
Epoch 2/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9396 - loss: 0.2027 - val_accuracy: 0.9451 - val_loss: 0.1860
Epoch 3/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 20ms/step - accuracy: 0.9533 - loss: 0.1357 - val_accuracy: 0.9451 - val_loss: 0.1489
Epoch 4/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - accuracy: 0.9698 - loss: 0.1000 - val_accuracy: 0.9670 - val_loss: 0.1325
Epoch 5/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 20ms/step - accuracy: 0.9808 - loss: 0.0805 - val_accuracy: 0.9670 - val_loss: 0.1210
Epoch 6/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - accuracy: 0.9835 - loss: 0.0672 - val_accuracy: 0.9670 - val_loss: 0.1178
Epoch 7/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 19ms/step - accuracy: 0.9863 - loss: 0.0580 - val_accuracy: 0.9560 - val_loss: 0.1153
Epoch 8/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 19ms/step - accuracy: 0.9863 - loss: 0.0522 - val_accuracy: 0.9560 - val_loss: 0.1131
Epoch 9/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.9835 - loss: 0.0453 - val_accuracy: 0.9560 - val_loss: 0.1085
Epoch 10/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.9890 - loss: 0.0417 - val_accuracy: 0.9451 - val_loss: 0.1078
Epoch 11/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9890 - loss: 0.0372 - val_accuracy: 0.9451 - val_loss: 0.1059
Epoch 12/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - accuracy: 0.9890 - loss: 0.0344 - val_accuracy: 0.9451 - val_loss: 0.1032
Epoch 13/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 16ms/step - accuracy: 0.9890 - loss: 0.0302 - val_accuracy: 0.9451 - val_loss: 0.1046
Epoch 14/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 21ms/step - accuracy: 0.9918 - loss: 0.0269 - val_accuracy: 0.9451 - val_loss: 0.1035
Epoch 15/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 23ms/step - accuracy: 0.9945 - loss: 0.0235 - val_accuracy: 0.9560 - val_loss: 0.1049
Epoch 16/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - accuracy: 0.9945 - loss: 0.0218 - val_accuracy: 0.9670 - val_loss: 0.1051
Epoch 17/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 30ms/step - accuracy: 0.9945 - loss: 0.0201 - val_accuracy: 0.9560 - val_loss: 0.1059
Epoch 18/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 24ms/step - accuracy: 0.9945 - loss: 0.0168 - val_accuracy: 0.9670 - val_loss: 0.1064
Epoch 19/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 33ms/step - accuracy: 0.9945 - loss: 0.0153 - val_accuracy: 0.9560 - val_loss: 0.1066
Epoch 20/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 31ms/step - accuracy: 0.9973 - loss: 0.0140 - val_accuracy: 0.9560 - val_loss: 0.1085
Epoch 21/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 36ms/step - accuracy: 0.9973 - loss: 0.0124 - val_accuracy: 0.9451 - val_loss: 0.1122
Epoch 22/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 16ms/step - accuracy: 0.9973 - loss: 0.0113 - val_accuracy: 0.9451 - val_loss: 0.1144
Epoch 23/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 26ms/step - accuracy: 0.9973 - loss: 0.0100 - val_accuracy: 0.9560 - val_loss: 0.1141
```

```
Epoch 25/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 13ms/step - accuracy: 1.0000 - loss: 0.0084 - val_accuracy: 0.9451 - val_loss: 0.1138
Epoch 26/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9973 - loss: 0.0077 - val_accuracy: 0.9451 - val_loss: 0.1200
Epoch 27/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 1.0000 - loss: 0.0069 - val_accuracy: 0.9451 - val_loss: 0.1234
Epoch 28/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 12ms/step - accuracy: 1.0000 - loss: 0.0067 - val_accuracy: 0.9451 - val_loss: 0.1188
Epoch 29/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 1s 17ms/step - accuracy: 1.0000 - loss: 0.0062 - val_accuracy: 0.9451 - val_loss: 0.1231
Epoch 30/30
23/23 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 1.0000 - loss: 0.0053 - val_accuracy: 0.9451 - val_loss: 0.1272
```

```
loss, accuracy = model.evaluate(X_test, y_test)

print("Test Accuracy:", accuracy)
```

```
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - accuracy: 0.9737 - loss: 0.0861
Test Accuracy: 0.9736841917037964
```

```
y_pred = model.predict(X_test)

y_pred = (y_pred > 0.5).astype(int)
```

```
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step
```

```
acc = accuracy_score(y_test, y_pred)

print("Accuracy =", acc)
```

```
Accuracy = 0.9736842105263158
```

```
cm = confusion_matrix(y_test, y_pred)

print(cm)
```

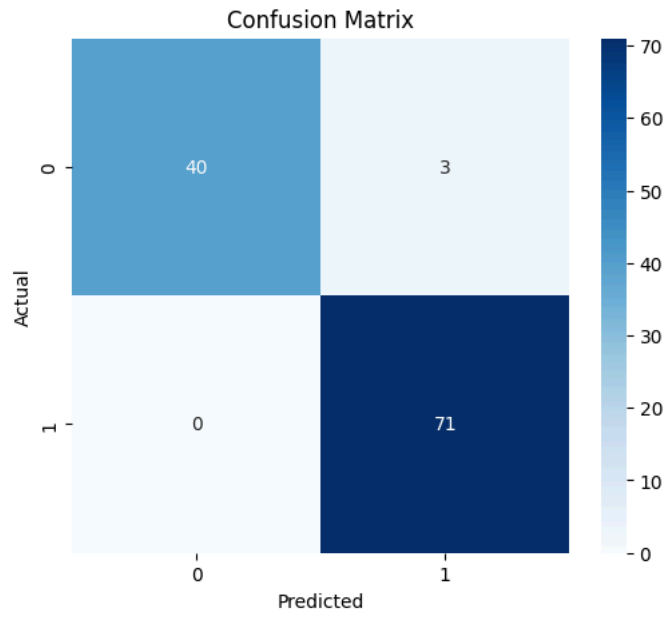
```
[[40  3]
 [ 0 71]]
```

```
plt.figure(figsize=(6,5))

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues'
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()
```




```
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       1.00      0.93      0.96        43
           1       0.96      1.00      0.98        71

    accuracy                           0.97       114
   macro avg       0.98      0.97      0.97       114
weighted avg       0.97      0.97      0.97       114
```

```
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')

plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.legend()

plt.show()
```

